

ISHARA HEWAWITHARANA

Computer Science Undergraduate (3rd Year)

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GitHub: github.com/ishara723

Portfolio: <https://ishara-hewawitharana.vercel.app>

PROFESSIONAL PROFILE

Undergraduate Computer Science student with a strong foundation in software development, programming, web technologies, and database management. Experienced in developing full-stack and backend applications using technologies such as Java, Spring Boot, React.js, MySQL, MongoDB, ASP.NET through academic & personal projects. Possesses strong problem-solving, analytical & teamwork skills, with a passion for learning new technologies & building innovative solutions. Seeking an internship opportunity to apply technical knowledge, gain hands-on industry experience & contribute effectively to real-world software development projects.

EDUCATION

BSc (Hons) in Computer Science

2024 - 2027

Sri Lanka Institute of Information Technology (SLIIT)

PROJECTS

Coordina – Community Collaboration & Resource Platform

[JAN 2026 – MAY 2026]

Technologies and Tools: React.js, ASP.NET Web API, ADO.NET, Supabase, Docker, Azure, GitHub Actions, SignalR

- Developed a full-stack community collaboration and resource management platform with role-based access for events, donation drives, bookings, tasks, and communication.
- Implemented synchronized multi-level calendar systems with real-time updates using SignalR and integrated booking, chat, and task management features.
- Built secure backend APIs with ASP.NET and Supabase, utilizing Docker and Azure deployment with CI/CD pipelines and comprehensive testing workflows.

Tropical Reserve Rescue (AI Wildlife Rescue Simulation)

[JAN 2026 – MAY 2026]

Technologies and Tools: Unity, C#, Blender, A* Pathfinding, BFS, Raycasting, Rigidbody Physics, LineRenderer

- Integrated physics-based obstacle removal, smooth drone animation, and visual debugging using Unity tools such as Rigidbody, Raycasting, and LineRenderer.
- Implemented A* pathfinding, BFS fallback navigation, and graph-based node systems for autonomous rescue drone navigation and real-time path recalculation.
- Developed an AI-powered 3D wildlife rescue simulation in Unity set in a tropical eco-tourism reserve with dynamic environmental interactions.

Distributed File Storage System

[JULY 2025 - NOV 2025]

Technologies and Tools: Python, Raft Consensus Algorithm, TCP Sockets, Vector Clocks, NTP

- Developed a fault-tolerant distributed storage system using the Raft consensus algorithm for leader election and log replication.
- Implemented real-time, heartbeat-based failure detection and a three-tiered consistency model (ONE, QUORUM, ALL) to ensure 99.2% availability.
- Utilized vector clocks for conflict resolution and TCP socket communication for authentic network connections between distributed storage nodes.

CIRO Mobile Phone Management System

[JAN 2025 - JUNE 2025]

Technologies and Tools: Spring Boot, MySQL, React, REST APIs

- Developed a full-stack mobile phone management system using Spring Boot, MySQL, and React.
- Implemented REST APIs and administrative CRUD operations for efficient management of phone records.

Sarasavi Library Management System

[JULY 2025 - NOV 2025]

Technologies and Tools: Java, Spring Boot, MongoDB, React.js

- Developed an automated digital solution to streamline library operations, including real-time tracking of books, members, and reservations.
- Implemented secure role-based access control (RBAC) for students and administrators using Spring Boot backend services.
- Built a responsive frontend interface with React.js and integrated it with a scalable MongoDB database.

Automated Waste Disposal System

[JAN 2024 - JUNE 2024]

Technologies and Tools: Raspberry Pi 3, OpenCV, Python, Servo Motors, Computer Vision Algorithms

- Developed an AI-driven automated waste segregation system utilizing computer vision to improve recycling efficiency.
- Integrated a Raspberry Pi with a camera module to capture real-time images and classify waste into categories like plastic, metal, and glass.
- Programmed motorized lid control using servo motors for accurate, contactless disposal based on object identification.

CERTIFICATIONS

2023: Computing For Career Development – University of Colombo School of Computing (UCSC)

2020: Web Designing Course – Zonal Computer Resource Centre, Taxila Central College

SKILLS

Technical Skills

Programming

Java, Python, C, JavaScript, C#

Web Development

HTML, CSS, Tailwind, React, Node.js, Spring Boot, ASP.NET

Database Management

MySQL, MongoDB, Supabase

Debugging & Testing

Postman, Selenium, Junit, JMeter

Version Control

GitHub

Methodologies

Agile, Scrum

Other

Figma, Blender, Balsamiq, Unity, Android Studio

Soft Skills

- Communication
- Teamwork
- Leadership
- Problem Solving
- Time Management
- Adaptability & Learning

REFERECES

Prof. Samantha Rajapaksha

Head/Department of IT
B.Sc. Eng. (Moratuwa), M.Sc. (SLIIT), Ph.D. (SLIIT).
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